Optimization of Capacity Allocation Using Linear Integer Programming in a Semiconductor Manufacturing Company

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ABSTRACT

This research addresses how to optimize capacity allocation in the photolithography area in a semiconductor manufacturing company using Linear Integer Programming (LIP). The main objective of this research is to optimize machine utilization rates and balance machine workloads across all machines while meeting the wafer demand. To tackle this problem, we develop an LIP model to optimize capacity allocation and determine the optimum wafer allocation quantities for each machine for each recipe. The study begins with a comprehensive explanation of semiconductor manufacturing, especially its significance and challenges in the photolithography area. The methodology includes problem identification, data collection, analysis, and LIP model development to be applied using MATLAB. Verification and validation process were conducted through step-by-step execution, sensitivity analysis, and scenario analysis. Results show that the optimization improved the average machine utilization from 88.87% to 69.46% while also reduced the Mean Absolute Deviation (MAD) from 43.23 to 0.003, indicating a more balanced workload. The study concludes with findings, contributions to semiconductor manufacturing field, and suggestions for future research.

Keywords: *Capacity Allocation, Linear Integer Programming (LIP), Photolithography, Semiconductor Manufacturing, Optimization*

1.0 INTRODUCTION

Semiconductor manufacturing, especially wafer-fabrication often known for its complex and high capital-intensive manufacturing that also involves complex processes in which each wafer goes through complicated sequence of process. This whole process of performing layer by layer on a single silicon wafer can take up to 700 repeating unit processing steps and consumed total minimum lead time of two months [1]. One of the crucial processes in semiconductor manufacturing is photolithography, where it takes 25%-40% of the total production cycle time [2] and has highly expensive tools complex resource constraints. Stepper and track machine – two of photolithography machines – are the most expensive and bottleneck machines in wafer fabrication site and most often have the highest utilization rate [3]. Nowadays, the cost for the most advanced step-and-scan sytems is over $50 million [7].

To meet the dynamic customer demand in time, utilizing and assigning machine capacity in a photolithography area properly is mandatory. While pursuing that target and make sure they don't surpass the machine utilization capacity, they also need to satisfy machine process capability (sometimes called as process window) constraint. Most past research termed this job as capacity allocation problem in photolithography area (CAPPA). However, in this research, we will be termed this problem as capacity allocation. According to Akcalt *et al.* (2000) study, it was demonstrated that a decreasing cycle time in photolithography process can lead to a reduced overall cycle time of fabrication. Hence, capacity allocation is essential in wafer fabrication process. Additionally, considering the significance production expenses and rapidly evolving technology, capacity allocation poses a significant challenge and essential for the investment and performance in the semiconductor industry. By allocating capacity properly, company in this industry can have a better competitive advantage in term of production cycle time and capital investment decision making for acquisition of machine equipment. This research aims to optimally distribute wafers allocation to photolithography machines by the development and implementation of LIP model to have an optimum machine utilization rates and balanced machine workloads while still be able to meet demand and not overutilizing any of the operating machines.

Currently, Company X allocates wafers based on decisions made by individual operators, leading to inefficiencies and suboptimal machine utilization. Operators often select work-in-progress (WIP) wafers to meet their daily key performance indicators (KPIs), without considering the overall planning for all machines. This lack of a systematic approach results in some machines being overloaded while others remain underutilized. The main issue identified from the current practice is the suboptimal wafers allocation, leading to overutilization of certain machines and unbalanced workloads.

Figure 1 shows the inefficiencies resulted from the current practice. The average machine utilization is 88.87%, indicating that many machines are working near or even pass their capacity limits, which reflects from the figure that there are certain machines that have machine utilization rates above 100%. Additionally, the Mean Absolute Deviation (MAD) of machine utilization is 43.23, showing that there is a huge imbalance in the workload distribution where some machines are heavily overutilized while some others are underutilized.

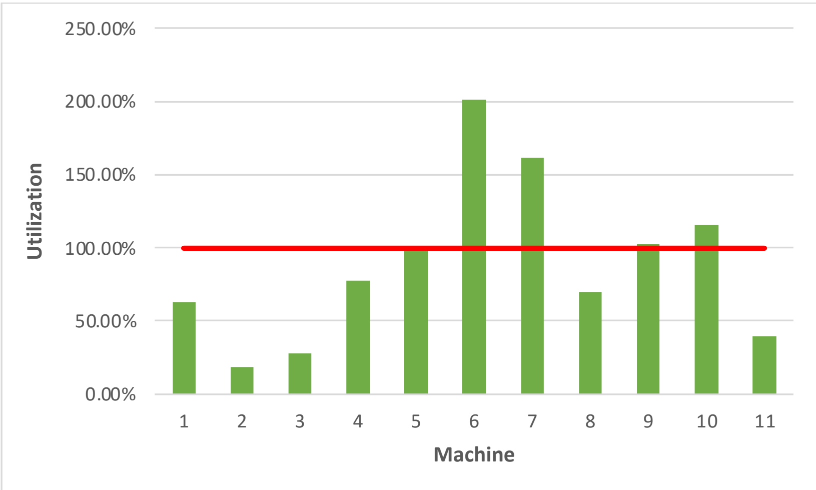

Figure 1: Current Practice's Machine Utilization Rates

2.0 LITERATURE REVIEW

Capacity allocation problem has been discussed a lot in previous research and each of it approached this problem differently. Ghasemi et al. (2020) proposed a new Mixed Integer

Linear Programming (MILP) model which combined with an improved genetic algorithm (GA). Next, Chen et al. (2016) employed a GA with an integer programming model and heuristic algorithm. Toktay and Uzsoy (1996) transformed the capacity allocation problem into maximum flow problems with integer-side constraints. Klemmt et al. (2010) approached this problem by a multistage MILP model on the basis of optimization approach for a planning problem in the photolithography area. Moreover, Akçalı and Uzsoy (2000) divided the shift-scheduling problem into two distinct sub-problems which are capacity allocation and lot sequencing, in order to analyze these in sequential manner. In another study, Chung et al. (2006) also introduced a MILP to address this capacity allocation problem. Chen et al. (2020) optimized capacity allocation by determine the optimal lotsplitting and lot-allocation with a mixed-integer nonlinear programming (MINLP) model combined with a hybrid genetic algorithm (HGA). Chung et al. (2008) solved the CAPPA by pointed out its NP-hard characteristic and used linear-based-programming heuristic algorithm (LBPHA). Ghasemi et al. (2018) used new genetic algorithm named RGA that derived from a psychological concept called Reference Group with a mixed integer programming formulation. Lastly, Chung et al. (2006) focused on CAPPA and presented the problem as part of Capacity Resource Planning (CRP) framework. By observing these previous research, we concluded that there are still some gaps that can be developed further through this research. This research will approach this capacity allocation problem by minimize the total maximum machine utilization rates among all machines and balance machine workloads distributions, without overutilizing any single machine by utilizing LIP model.

3.0 METHODOLOGY

This research's methodology started with the problem identification of capacity allocation problem in a semiconductor manufacturing company in Malaysia, including the collection of necessary data, such as machine process capabilities information, tool processing times, machine availability details, and demand quantities. These data are then analyzed in order to understand the problem better and to develop the suitable LIP model. The developed model is then applied and solved Optimization Toolbox in MATLAB software. Lastly, we conducted verification and validation process with step-by-step execution, sensitivity analysis, and scenario analysis to ensure the accuracy and adaptability of our model and results.

4.0 MODEL DEVELOPMENT

This research propose an LIP model aims to address this problem by optimizing wafer allocation to minimize the maximum machine utilization rates and balance machine workloads.

The LIP model is represented as:

$$\textbf{Minimize } \boldsymbol{Z = Maximum(\sum_{j=1}^{r} U_{1j}, \sum_{j=1}^{r} U_{2j}, \sum_{j=1}^{r} U_{3j}, \sum_{j=1}^{r} U_{4j}, \sum_{j=1}^{r} U_{5j}, \sum_{j=1}^{r} U_{6j}, \sum_{j=1}^{r} U_{7j}, \sum_{j=1}^{r} U_{8j}, \sum_{j=1}^{r} U_{9j}, \sum_{j=1}^{r} U_{10j}, \sum_{j=1}^{r} U_{11j})} \quad (1)$$

Subject to

$$\boldsymbol{\sum_{j=1}^{r} U_{ij} \leq 100\%} \ \forall \boldsymbol{i} \quad (2)$$
$$\boldsymbol{\sum_{i=1}^{n} x_{ij} = D_j} \quad \forall \boldsymbol{j} \quad (3)$$
$$\boldsymbol{mpc_{ij} = [0,1]} \quad (4)$$
$$\boldsymbol{x_{ij} \in Z^{+}} \quad \forall \boldsymbol{i,j} \quad (5)$$

$$\boldsymbol{x_{ij} \geq 0 \qquad \forall i,j} \tag{6}$$

Where x_{ij} represents represent the number of wafers allocated to machine i for recipe j, $U_{ij} = \frac{x_{ij} * p_{ij}}{t_{ij}} * 100\%$ and it represents the utilization rate of machine i for recipe j, p_{ij} is the value of tool processing time of machine i for recipe j, mpc_{ij} represents the machine process capability constraint of machine i for recipe j, t_{ij} is the value of machine availability of machine i for recipe j, n is the number of machines, r is the number of recipes, and D_j represent the demand for each recipe.

In the LIP model, Equation (1) is the objective function that focus to minimize the highest total machine utilization rates across all machines, from machine 1 to machine 11. Equation (2) is the capacity constraint to ensure that the total machine utilization rate for each machine does not exceed their capacity limit, which is 100%. Equation (3) represents the demand constraint which will make sure that the total wafer allocations for each recipe meets the exact total demand for that recipe. Equation (4) is the machine process capability constraint to determine whether a machine is capable to process a particular recipe or not. Equation (5) is the integer constraint to specify that the decision variables must be positive integers value. Lastly, Equation (6) is the non-negativity constraint to ensures that the decision variables value are not negative value.

5.0 RESULTS AND DISCUSSION

The application of the LIP model has significantly improved the capacity allocation in the photolithography area. The complete results of the detailed wafer allocations to each machine for each recipe can be seen in Table 1.

Table 1: Wafer Allocations to Each Machine for Each Recipe after Optimization

		MACHINE										
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**
RECIPE	**A**	9500										
	B	1286				8064						
	C	8695										
	D	9					6531					
	E			5645								
	F	30		6788						2		
	G							5346			4514	
	H		14531	2		2						
	I				8067			3				4470
	J	3760			4121		1215	1594				

	K		3959							4731		
	L	2666	5431						5903			

After applying the LIP model, the average machine utilization rates improved significantly, dropping to 69.46%. This improvement highlights that there is a more efficient used of the machines, redistributing the burdens from overutilized machines to those with more spare capacity. Moreover, the MAD value is also drastically reduced to just 0.003. This significant decrease indicates the model successfully uniform the distribution of workloads across all machines. The illustration of the optimized machine utilization rates and balanced workloads distribution can be seen in Figure 2.

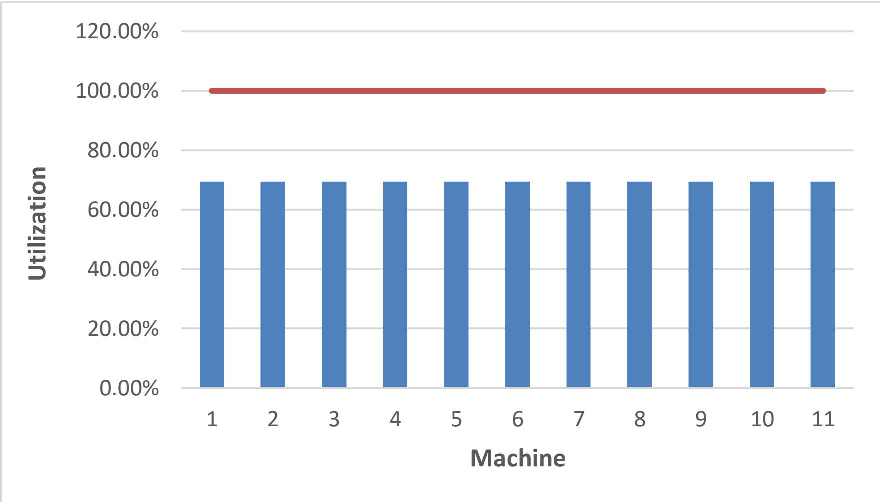

Figure 2: Optimized Machine Utilization Rates

This research's results show that our LIP model has been effective to optimize the capacity allocation in the photolithography area. This shown by a more balanced machine workloads distribution and reduced average machine utilization and MAD value. This improvement may lead to several positive impacts. Firstly, by balancing distribution of machine workloads, it will reduce the risk of overutilization in any machines that eventually will extend machines' lifespan because the potential of maintenance needs and machine breakdowns are being decreased. Next, the company can handle potential increase in demand in the future without having to acquire additional machines because now they have a lower average of machine utilization rates and more space in their machines to process additional demands. This will help the company to avoid huge potential capital expenditure. Lastly, this improvement will lead to a higher productivity due to a more stable and predictable production process as it will help the company to schedule their production better and reduce bottlenecks.

6.0 CONCLUSION

This research has been successful to develop and apply an LIP model to optimize the capacity allocation in a semiconductor manufacturing company by determined the optimum wafer allocation quantities for each machine and each recipe. As a result, it has optimized the machine utilization rates and balanced the distribution of machine workloads. These results have contributed to both theoretical knowledge and practical applications of this semiconductor manufacturing field. Future research may explore the best way to integrate real-time production data to the model and expand the model's application to not only other process in the semiconductor manufacturing, but also to other industries. These suggestions will improve the model's accuracy and adaptability.

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